

```

In [2]: # =====
# Assignment 8 - Exercise 1
# Unsupervised Learning on Titanic Dataset
# Data Exploration and Visualization
# =====

import pandas as pd
import matplotlib.pyplot as plt

# -----
# Step 1 : Load Dataset
# -----

titanic = pd.read_csv(r"C:\Users\91953\titanic.csv")

print("="*60)
print("Original Dataset")
print("="*60)
print(titanic.head())

# -----
# Step 2 : Remove Target Column
# -----

if 'Survived' in titanic.columns:
    unsupervised_data = titanic.drop(columns=['Survived'])
else:
    unsupervised_data = titanic.copy()

print("\n" + "="*60)
print("Dataset After Removing 'Survived' Column")
print("="*60)
print(unsupervised_data.head())

# -----
# Step 3 : Dataset Information
# -----

print("\n" + "="*60)
print("Shape of Dataset")
print("="*60)
print(unsupervised_data.shape)

print("\n" + "="*60)
print("Column Names")
print("="*60)
print(unsupervised_data.columns.tolist())

print("\n" + "="*60)
print("Missing Values")
print("="*60)
print(unsupervised_data.isnull().sum())

# -----
# Step 4 : Age Distribution
# -----

plt.figure(figsize=(8,5))

```

```
plt.hist(
    unsupervised_data['Age'].dropna(),
    bins=20,
    edgecolor='black'
)

plt.title("Distribution of Passenger Age")
plt.xlabel("Age")
plt.ylabel("Number of Passengers")
plt.grid(alpha=0.3)

plt.show()

# -----
# Step 5 : Fare Distribution
# -----

plt.figure(figsize=(8,5))

plt.hist(
    unsupervised_data['Fare'].dropna(),
    bins=20,
    edgecolor='black'
)

plt.title("Distribution of Passenger Fare")
plt.xlabel("Fare")
plt.ylabel("Number of Passengers")
plt.grid(alpha=0.3)

plt.show()

# -----
# Step 6 : Passenger Class Distribution
# -----

plt.figure(figsize=(7,5))

unsupervised_data['Pclass'].value_counts().sort_index().plot(
    kind='bar',
    edgecolor='black'
)

plt.title("Passenger Class Distribution")
plt.xlabel("Passenger Class")
plt.ylabel("Number of Passengers")
plt.grid(axis='y', alpha=0.3)

plt.show()

# -----
# Step 7 : Observations
# -----

print("\n" + "="*60)
print("Observations")
print("="*60)

print("1. Most passengers are adults between 20 and 40 years of age.")
```

```
print("2. Passenger fares are right-skewed, with most passengers paying lower fa  
print("3. Passenger Class 3 has the highest number of passengers.")  
print("4. The dataset is now ready for Unsupervised Learning algorithms such as  
print("\nExercise 1 Completed Successfully.")
```

```
=====
Original Dataset
=====
```

	Passengerid	Age	Fare	Sex	sibsp	zero	zero.1	zero.2	zero.3	\
0	1	22.0	7.2500	0	1	0	0	0	0	
1	2	38.0	71.2833	1	1	0	0	0	0	
2	3	26.0	7.9250	1	0	0	0	0	0	
3	4	35.0	53.1000	1	1	0	0	0	0	
4	5	35.0	8.0500	0	0	0	0	0	0	

	zero.4	...	zero.12	zero.13	zero.14	Pclass	zero.15	zero.16	Embarked	\
0	0	...	0	0	0	3	0	0	2.0	
1	0	...	0	0	0	1	0	0	0.0	
2	0	...	0	0	0	3	0	0	2.0	
3	0	...	0	0	0	1	0	0	2.0	
4	0	...	0	0	0	3	0	0	2.0	

	zero.17	zero.18	Survived
0	0	0	0
1	0	0	1
2	0	0	1
3	0	0	1
4	0	0	0

```
[5 rows x 28 columns]
```

```
=====
Dataset After Removing 'Survived' Column
=====
```

	Passengerid	Age	Fare	Sex	sibsp	zero	zero.1	zero.2	zero.3	\
0	1	22.0	7.2500	0	1	0	0	0	0	
1	2	38.0	71.2833	1	1	0	0	0	0	
2	3	26.0	7.9250	1	0	0	0	0	0	
3	4	35.0	53.1000	1	1	0	0	0	0	
4	5	35.0	8.0500	0	0	0	0	0	0	

	zero.4	...	zero.11	zero.12	zero.13	zero.14	Pclass	zero.15	zero.16	\
0	0	...	0	0	0	0	3	0	0	
1	0	...	0	0	0	0	1	0	0	
2	0	...	0	0	0	0	3	0	0	
3	0	...	0	0	0	0	1	0	0	
4	0	...	0	0	0	0	3	0	0	

	Embarked	zero.17	zero.18
0	2.0	0	0
1	0.0	0	0
2	2.0	0	0
3	2.0	0	0
4	2.0	0	0

```
[5 rows x 27 columns]
```

```
=====
Shape of Dataset
=====
```

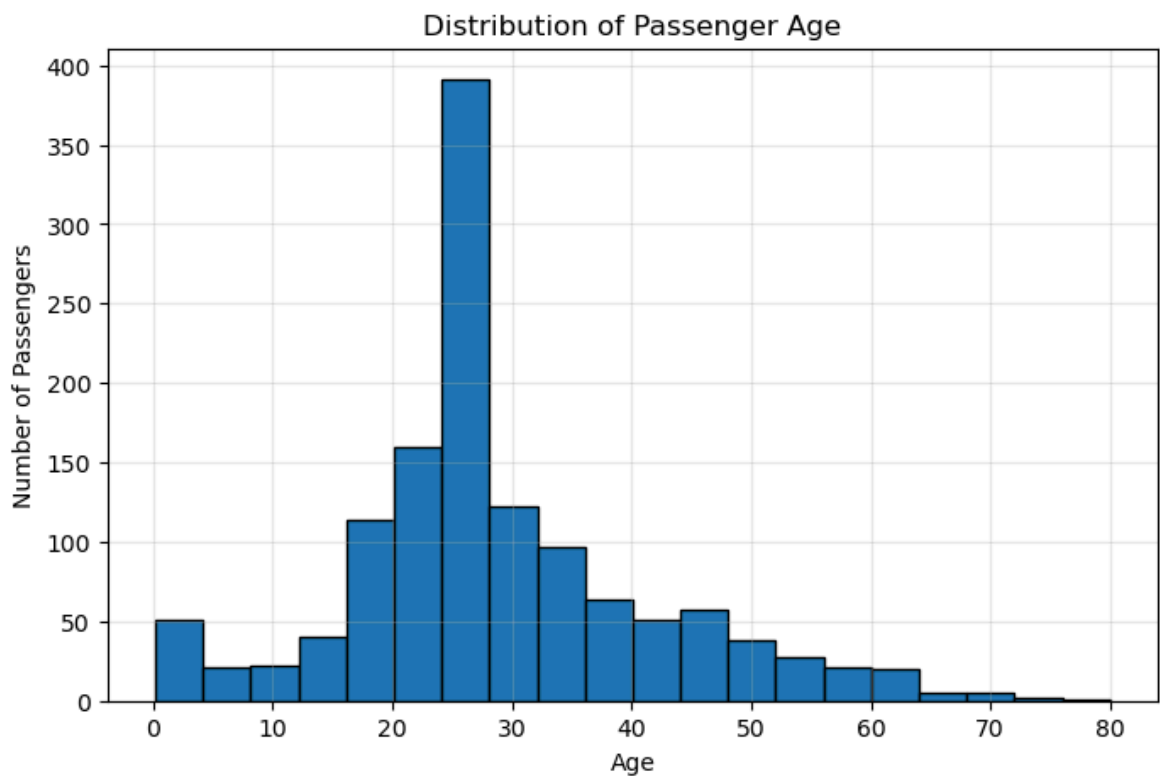
```
(1309, 27)
```

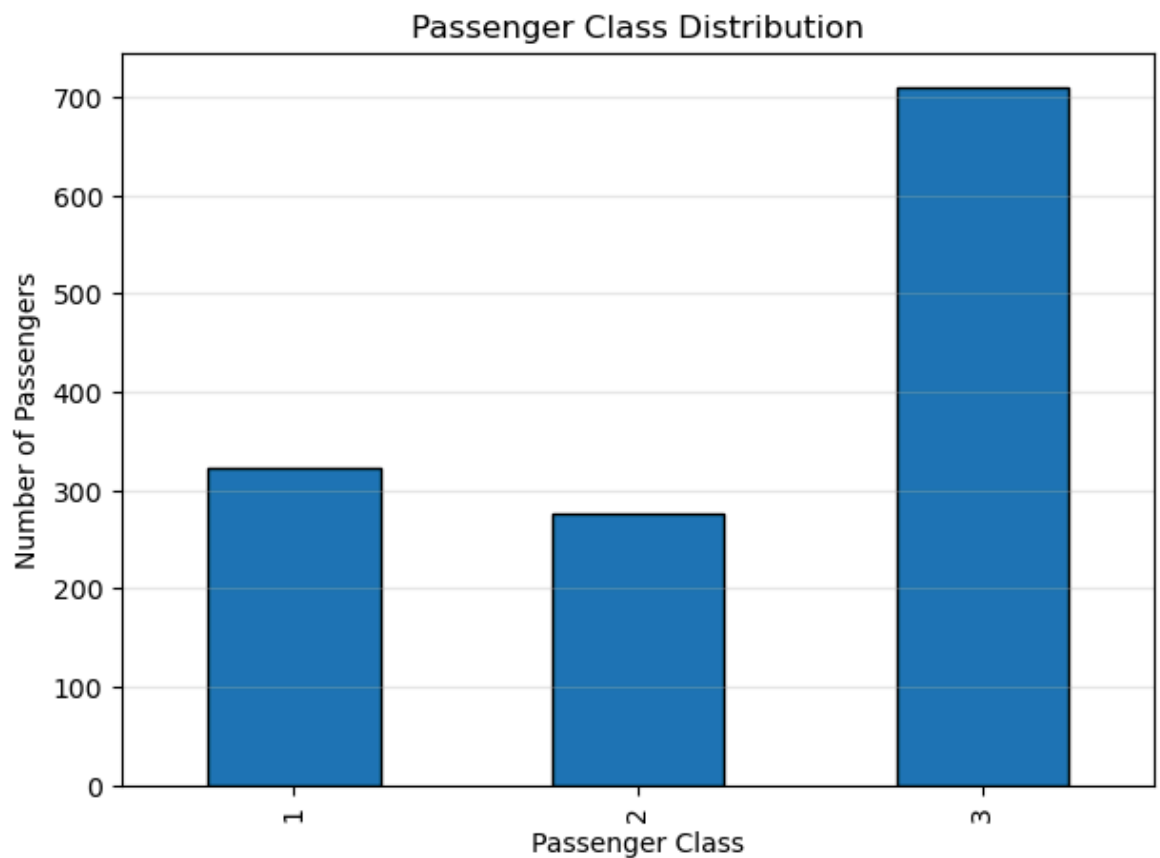
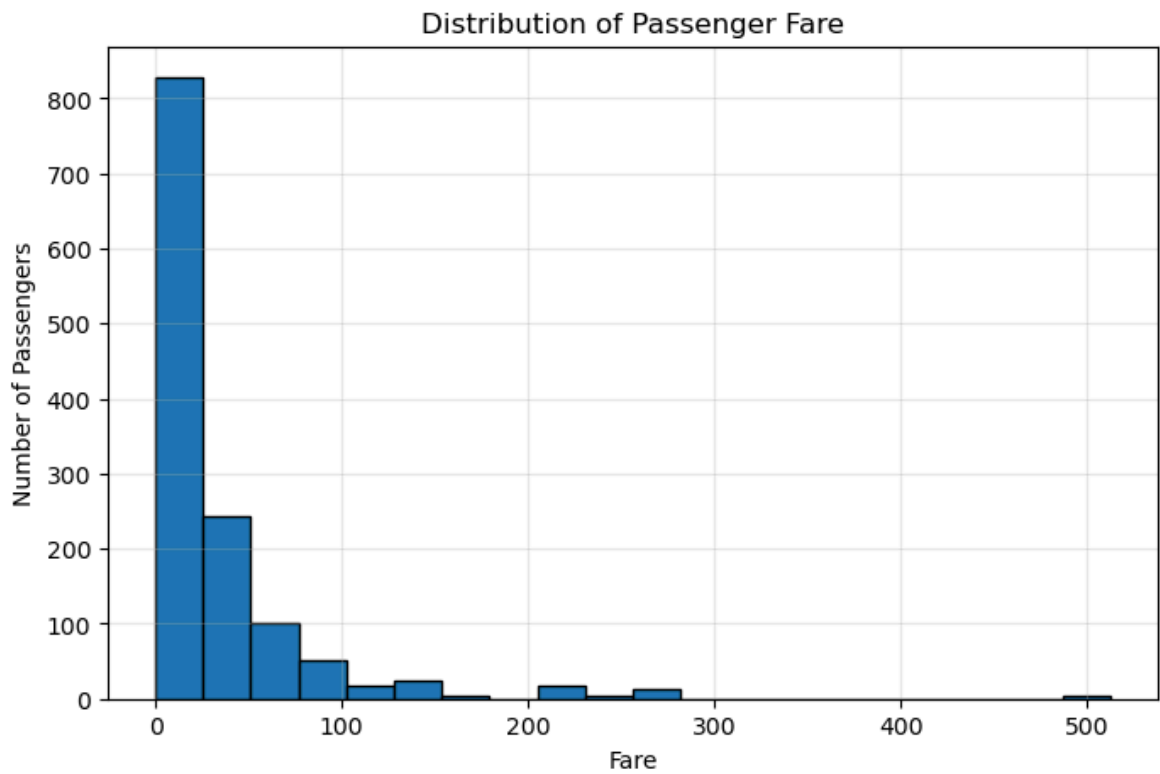
```
=====
Column Names
=====
```

```
['Passengerid', 'Age', 'Fare', 'Sex', 'sibsp', 'zero', 'zero.1', 'zero.2', 'zero.3', 'zero.4', 'zero.5', 'zero.6', 'Parch', 'zero.7', 'zero.8', 'zero.9', 'zero.10', 'zero.11', 'zero.12', 'zero.13', 'zero.14', 'Pclass', 'zero.15', 'zero.16', 'Embarked', 'zero.17', 'zero.18']
```

```
=====
Missing Values
=====
```

```
Passengerid    0
Age             0
Fare           0
Sex            0
sibsp          0
zero           0
zero.1         0
zero.2         0
zero.3         0
zero.4         0
zero.5         0
zero.6         0
Parch          0
zero.7         0
zero.8         0
zero.9         0
zero.10        0
zero.11        0
zero.12        0
zero.13        0
zero.14        0
Pclass         0
zero.15        0
zero.16        0
Embarked       2
zero.17        0
zero.18        0
dtype: int64
```





=====
Observations

=====

1. Most passengers are adults between 20 and 40 years of age.
2. Passenger fares are right-skewed, with most passengers paying lower fares.
3. Passenger Class 3 has the highest number of passengers.
4. The dataset is now ready for Unsupervised Learning algorithms such as K-Means Clustering.

Exercise 1 Completed Successfully.

In []: